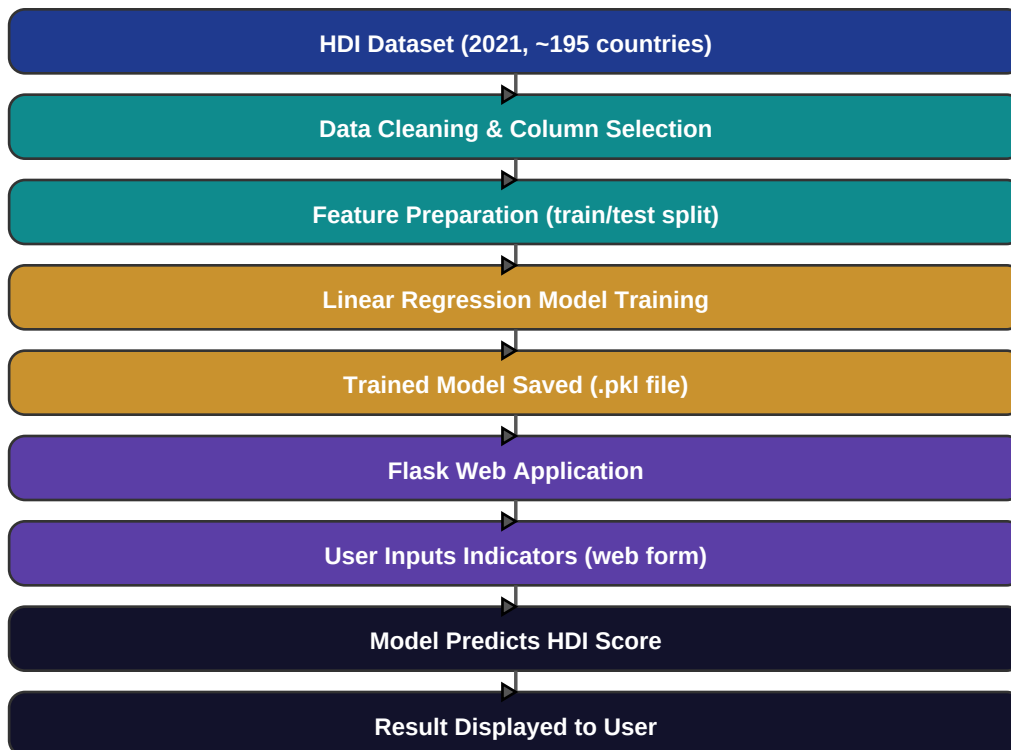


Data Flow Diagram

HDI Prediction System | Team Lead: Farzana | Contributors: Chinthoti Raghavendra, Ahmad Kabir Shaik, Naga Sireesha Naralasetty, Sitha Ramaraju Velisala

01 System overview

The diagram below shows how data moves through the HDI Prediction System — from raw dataset collection to the final predicted HDI score shown to the user.



02 Flow description

- Raw HDI dataset (2021 indicators for ~195 countries) is loaded and cleaned, keeping only relevant columns.
- Data is split into training and testing sets to prepare for model training.
- A Linear Regression model is trained on the processed data, achieving an R^2 score of approximately 0.98.
- The trained model is saved as a .pkl file and loaded into a Flask web application.
- Users submit indicator values through a web form, which the Flask app passes to the model.
- The model returns a predicted HDI score, which is displayed back to the user instantly.